

Question 1: We study the following SIR model without demography:

$$\frac{dS}{dt} = -\frac{\beta SI}{N}, \quad \frac{dI}{dt} = \frac{\beta SI}{N} - \gamma I, \quad \frac{dR}{dt} = \gamma I, \quad (1)$$

with contact rate, $\beta = 0.62$ (days⁻¹), and a mean infectious duration of 2.6 (days). This implies the recovery rate is the reciprocal of the mean infectious duration, hence $\gamma = 1/26$. We assume a population size of $N = 62,000,000$.

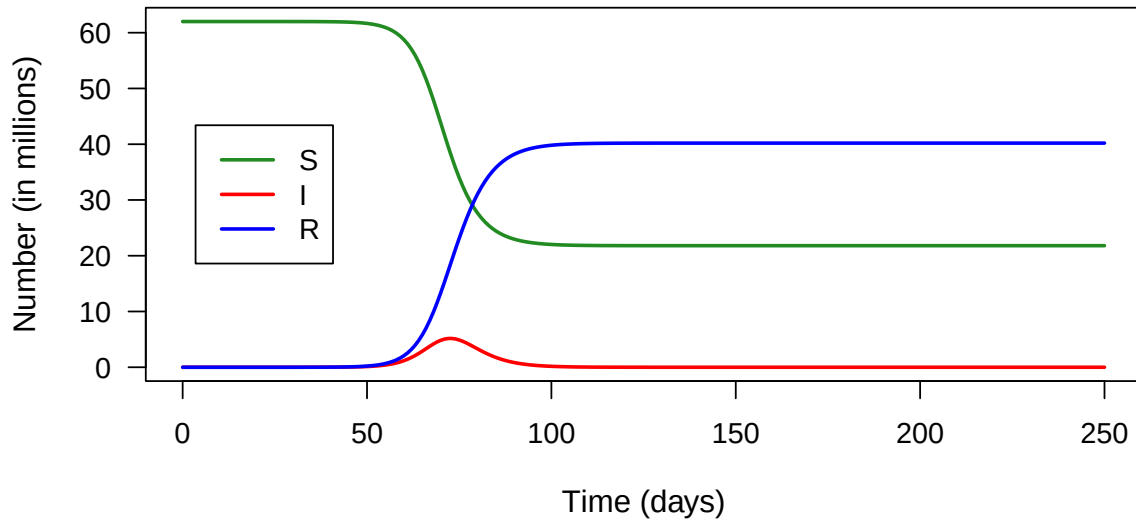


Figure 1: Dynamics of all SIR model classes over time.

We use R code, specifically the line `FinalSize <- round(Classes$R[length(Classes$R)])`, to compute the final size of the epidemic, which yields $R_\infty = 40,198,772$. From lectures, we know for this SIR model $R_0 = \frac{\beta}{\gamma} = \frac{62 \cdot 2.6}{100} = 1.612$. From our R code, we deduce that the expected duration is 172 days, where expected duration is defined as the time during which there is at least 1 infection present.

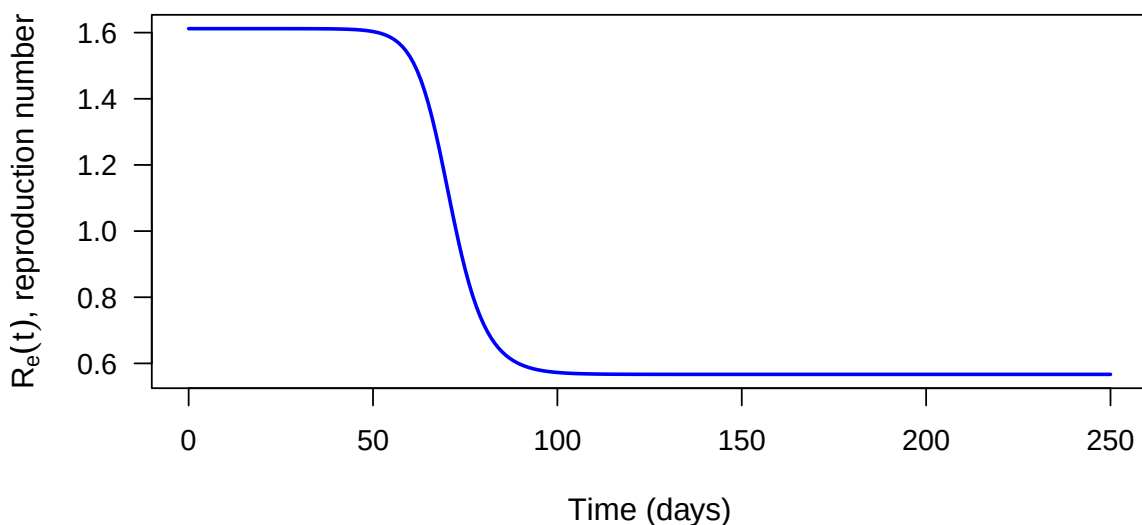


Figure 2: Effective reproduction number, $R_e(t)$, over time.

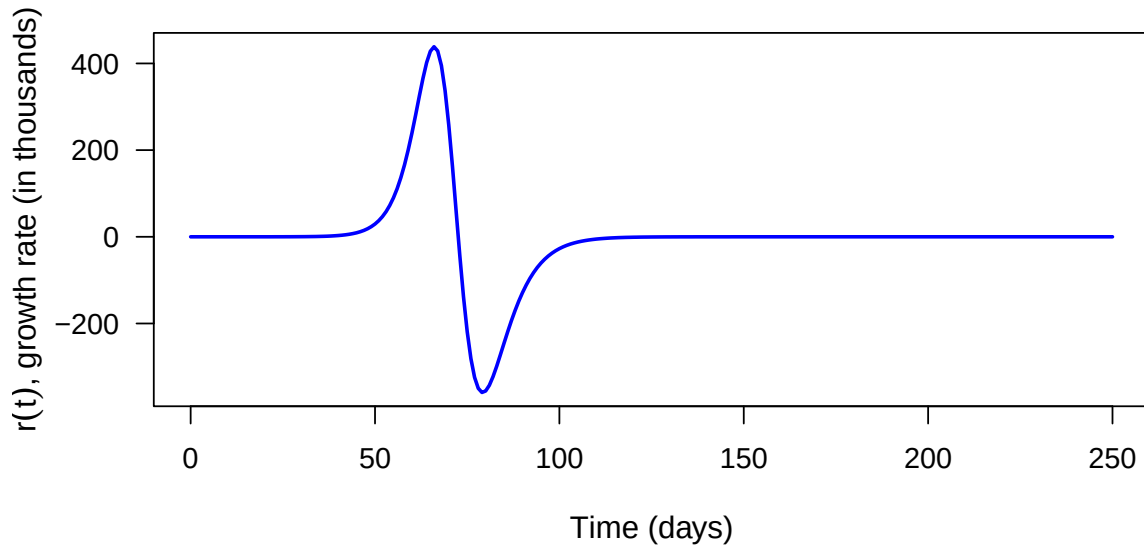


Figure 3: Growth rate, $r(t)$, over time.

The peak of the infection happens when $\frac{dI}{dt} = 0$. Therefore, we have $\frac{\beta S_{\text{peak}} I}{N} = \gamma I$, which implies $S_{\text{peak}} = \frac{N\gamma}{\beta} = \frac{62,000,000}{0.62 \cdot 2.6} = 38,461,539$, to the nearest integer. Here, S_{peak} denotes the number of susceptibles at peak infection.

Question 2: We extend our model to now have two infection classes namely I_r , representing infected individuals **with** symptoms severe enough to be *reported*, and I_n , representing those **without** symptoms severe enough to be reported. We will assume that, due to reporting delays and the short infection period, cases are reported on the day the infected person recovers. So our model now looks like:

$$\begin{aligned} \frac{dS}{dt} &= -\frac{\beta S(I_r + I_n)}{N}, & \frac{dI_r}{dt} &= \frac{p\beta S(I_r + I_n)}{N} - \gamma I_r & (2) \\ \frac{dI_n}{dt} &= \frac{(1-p)\beta S(I_r + I_n)}{N} - \gamma I_n, & \frac{dR}{dt} &= \gamma(I_r + I_n). & (3) \end{aligned}$$

The new parameter p represents the proportion of infected individuals with symptoms severe enough to be reported. This is deduced from the $\frac{dI_r}{dt}$ component, $\frac{\beta S(I_r + I_n)}{N}$ describes the total rate at which susceptible individuals become infected due to contact with other infected individuals (both reported and unreported). Multiplying this by p accounts for the fraction of these newly infected individuals who develop symptoms severe enough to be reported.

We assume $p = 0.04$. If we assume 13% of the population already have prior immunity when the first infection was introduced to the UK and further assume that this first infection does not cause symptoms severe enough to be reported, then our model has initial conditions of

$$(S^*, I_r^*, I_n^*, R^*) = (N - R^* - I_r^* - I_n^*, 0, 1, 0.13 \cdot N) = (53939999, 0, 1, 8060000). \quad (4)$$

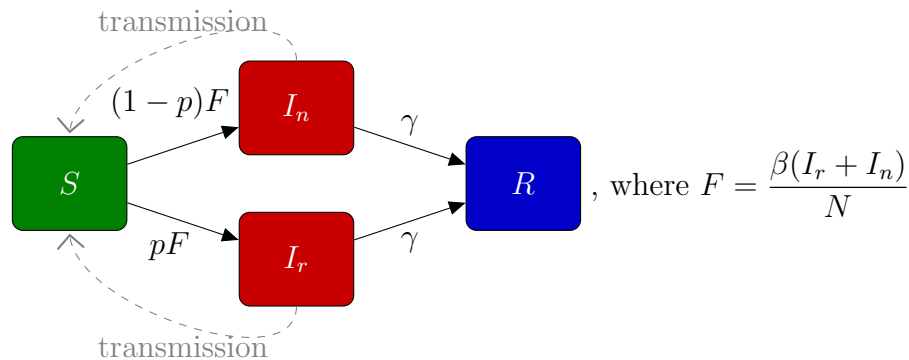


Figure 4: Modified SIR model compartmental diagram

In this model, the current infection prevalence is given by $I_r(t) + I_n(t)$, as a function of time. The daily reported infection incidence across the 250-day period is given by $\gamma I_r(t)$, as a function of time. This is because firstly, the model assumes that we only report cases that are severe, stored in I_r . Secondly, we also assume that cases are reported on the day the infected individual has recovered, therefore we are interested in the transition between classes I_r to R . This rate of change is governed by the $\gamma(I_r + I_n)$ term as seen in the model, but we recall only individuals from the I_r class are reported and therefore we ignore the presence of I_n . Since we already compute this on a daily interval, we do not need to integrate this force of recovery due to I_r . This yields the aforementioned formula.

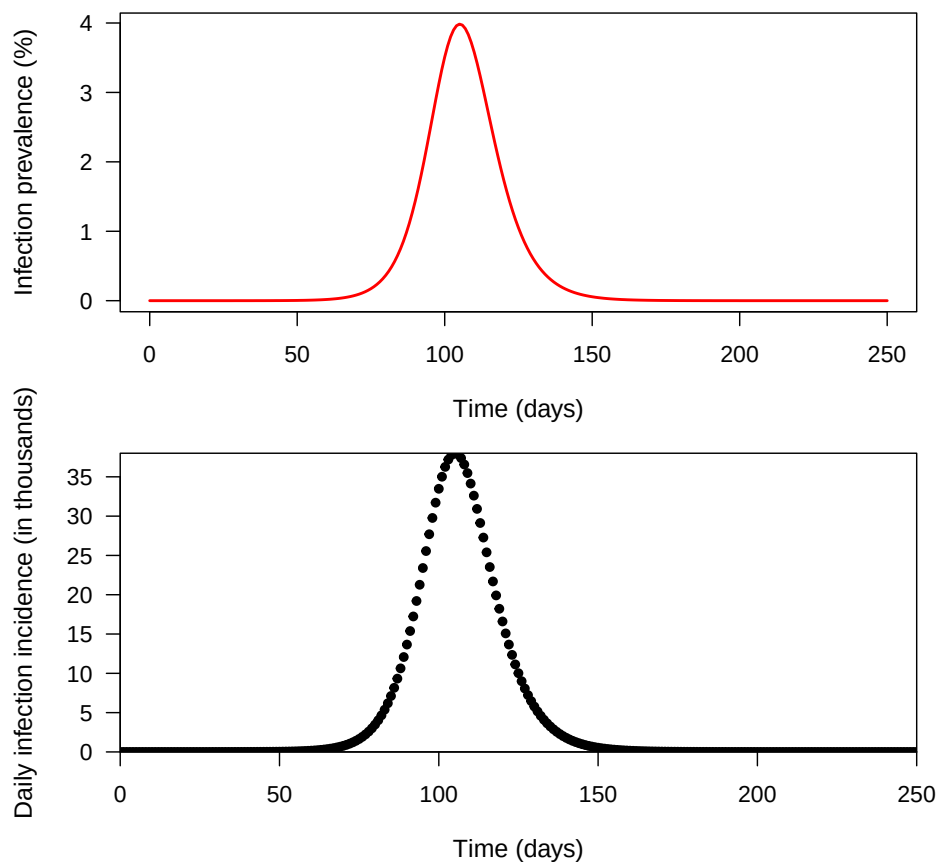


Figure 5: Current Infection Prevalence (top) and daily reported infection incidence.

By calling the `trapz` function to sum up the number of reported infections, we obtain that the cumulative count of reported infections was 1,106,719. Based on our model, the expected number of people with immunity is R_∞ , since we assume immunity when the infected individual has recovered. Using our R code, we find $R_\infty = 35,727,953$.

Question 3: Some key differences between our model outputs and the bar plot are:

1. Our model output, specifically Figure (9), has more granular data, showing daily variations in infection incidence where as the bar plot shows weekly variations in infection incidence.
2. Our model output has only peak, where as the bar plot shows two peaks suggesting the presence of a secondary infection wave. One reason for this is because our model doesn't account for the possibility that recovered people could get infected again. Our model assumes the probability of immunity after infection is 1.0 and that immunity lasts forever, but realistically this is not the case.
3. Our model output is very smooth unlike the bar plot which seems to have some form of noise. This is because in reality, there could be a bias in reported cases during different times of the duration period, but we assume a perfect reporting pattern in our model.
4. Due to the different granularity of our incidence data and the bar plots incidence data, we also have different peak sizes.

We run our original model again but we introduce the concept of *school holidays* whereby we assume the infection was introduced in epidemiological week 17; the school holidays occur between weeks 30 to 35 inclusive; and during these holidays, there was a contact rate reduction of 60%. The infection prevalence dynamics are shown below:

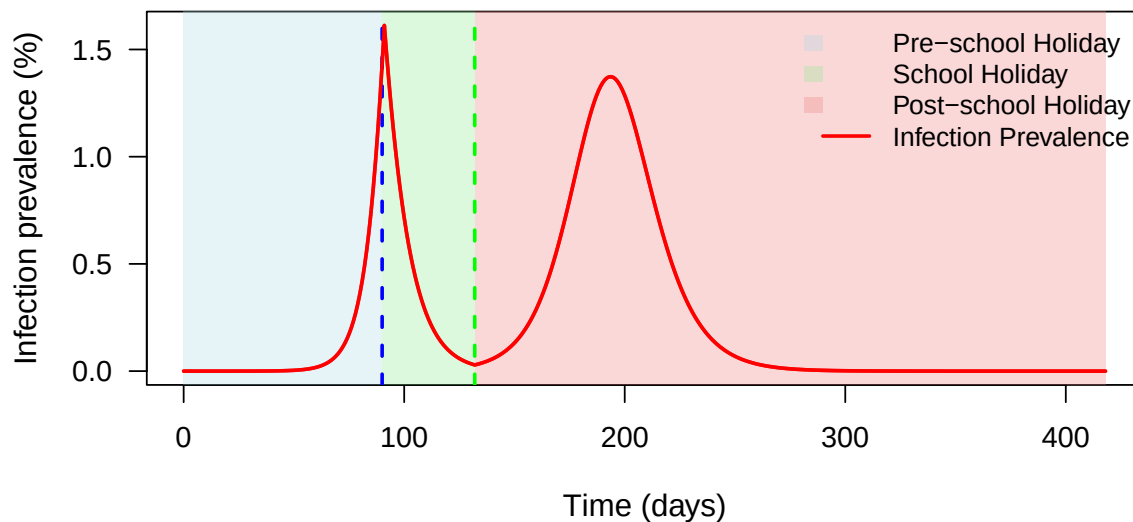


Figure 6: Infection prevalence (%), over 59 week period.

During the pre-school holiday phase, $R_e(t)$ behaves like before since the model simulates exactly as before. Then at the start of the school holiday phase, we introduce a reduction factor of 0.6 to β which reduces contact rates so we expect $R_e(t)$ to suddenly decline as it does. Then at the start of the post-school holiday phase, we fix β back to

its original value which increases contact rate so we expect $R_e(t)$ to suddenly increase as it does and then it follows the shape of the original model. The sharp changes arise from introducing and removing the reduction factor instantaneously, in reality it should be done in a smoother manner.

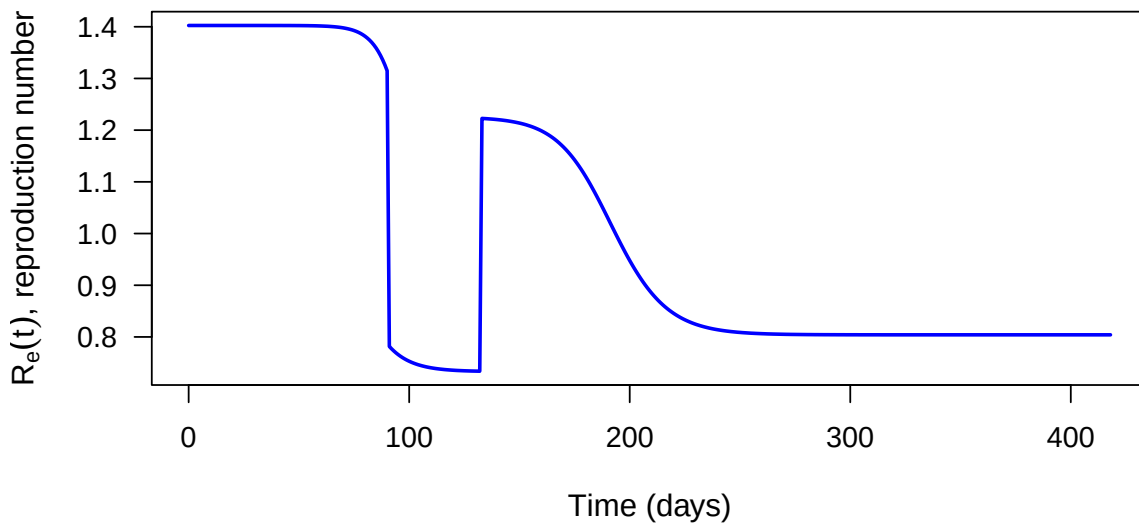


Figure 7: Effective reproduction number, $R_e(t)$, over 59 week period.

The final size of the outbreak is 31,074,042 and the expected duration of the outbreak is 391 days or 55.86 weeks.

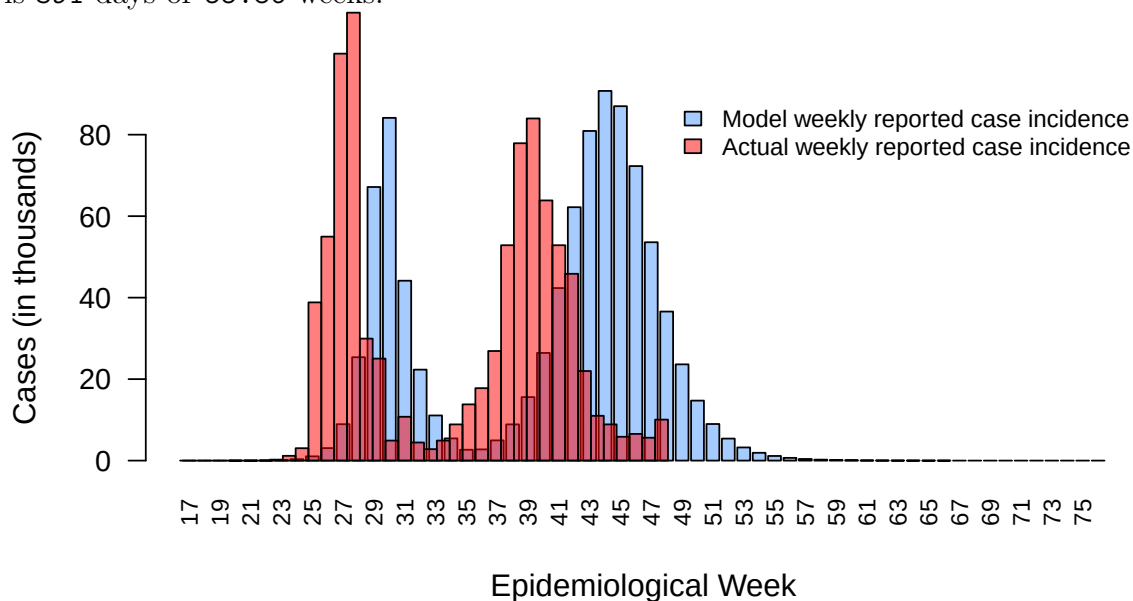


Figure 8: Weekly reported case incidence; actual data against model result.

Present in both our simulation and the actual data are two distinct peaks which represent the outbreak prior and post school holidays. The general shape of incidence data is similar between the two data sets but there is a clear lag time between our model and the actual data. This is most likely because in our model, we assume reporting occurs once the infected individual has recovered, but in reality reports would be submitted much earlier.

Question 4: The Herd Immunity Threshold (HIT) is calculated as $1 - 1/R_0$, which in our case equals $1 - 1/1.612 = 0.378$. Thus, 37.8% of the population requires immunity

to significantly slow flu spread. Our model assumes recovered individuals gain immunity, with a final size of 31,074,042 (50.1% of the population). This exceeds the HIT, creating an overshooting model where $R_e(t)$ eventually drops below 1, slowing but not immediately halting the epidemic. Adding immunity factors like vaccination could limit overshoot. Similarly, school holidays reduce transmission rates, helping to slow the epidemic and limit overshoot.

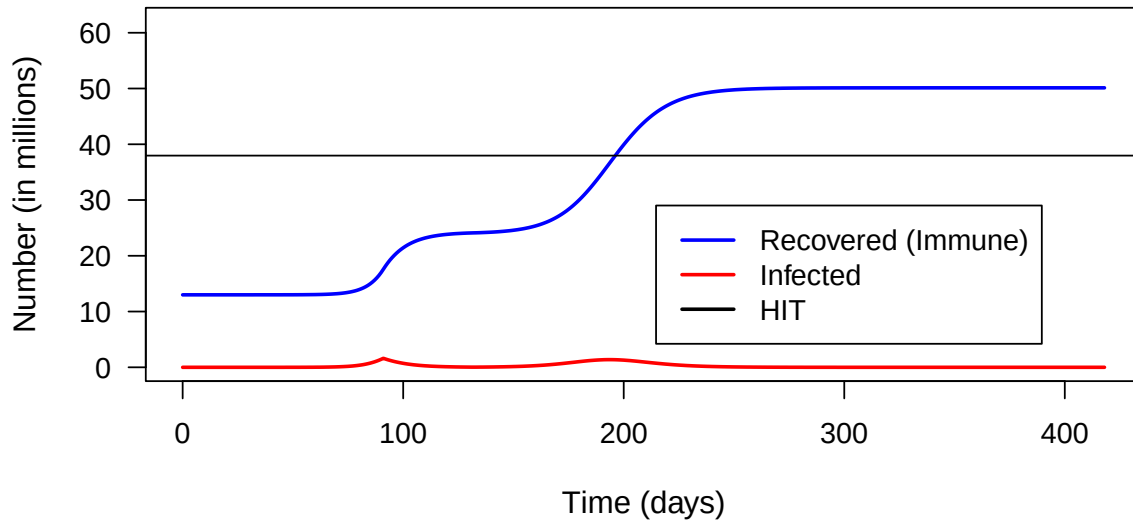


Figure 9: Proportion of immune people compared to HIT, demonstrating overshoot.

The increase in recoveries beyond the HIT illustrates overshoot in our model. Additionally, we observe two distinct sudden increases in recoveries: one driven by the impact of school holidays reducing transmission, and another by the accumulation of immunity in the population. The presence of overshoot suggests there are ways to reduce the rate of infection with school holidays being an example. Any infections beyond the time where our HIT curve intersects the recovered curve is due to overshoot. Another way to compute overshoot is $R_\infty - R_{\text{peak}}$ which effectively computes infections after hitting the peak. For Q1 this is 20,970,784, for Q2 this is 14,766,713 and for Q3 this is 8,242,873. We can observe by introducing prior immunity and school holidays, we successfully reduce the overshoot value across the models.

Aiming for herd immunity through natural infection requires a large portion of the population to become infected. Our model doesn't consider age, meaning vulnerable groups like the elderly and children, who are more likely to experience severe illness or death, are not accounted for. Allowing natural infection would be socially inequitable, as some people may suffer disproportionately due to their age or health status. Furthermore, socioeconomic differences may prevent access to adequate healthcare for certain individuals, exacerbating the social inequity of this approach. The model assumes everyone is equally affected by the infection, which is not true in reality. Individuals with conditions like asthma may face worsened symptoms, reducing their quality of life, while those with weaker immune systems are more likely to suffer. This undermines the concept of social equity. Additionally, the model does not consider that immune response times vary among individuals. People with stronger immune systems are likely to recover faster, leaving those with weaker immune systems at greater risk. These disparities highlight the importance of considering health inequalities and immune differences, which are not captured by the current model.